# Minki Lee

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Integrated M.S. + Ph.D. Program Student
CompSec Lab [https://compsec.postech.ac.kr],
Department of Computer Science and Engineering,
Pohang University of Science and Technology (POSTECH), South Korea

## Interests

- {Cyber-Physical Systems (CPS) | Distributed Systems | Embedded Systems} Security
- Secure-by-Design Development
- CPS Architecture & Engineering
- Open Source Software Security

## Skills

- **Languages**: Korean (Native), English (Proficient)
- **Computer Languages**: C/C++, Python, JavaScript, PHP, Dart
- **Tools and Testbeds**: PX4/Pixhawk system, ROS2, gazebo, Crazyflie, Matlab simulations

## Education

**Integrated M.S. & Ph.D. Program** — September 2025 - Present
Computer Security Lab, POSTECH CSE — *Pohang, South Korea*

- Advisor: Seulbae Kim
- Research in Cyber-Physical Systems' security
- Swarm system resilience and robustness

**Bachelor of Engineering in Computer Science Major** — February 2021 - August 2025
DGIST, School of Undergraduate Studies — *Daegu, South Korea*

- WAM/GPA: 3.60/4.30, percent scale: 93.0%
- Bachelor's Thesis-substitute Report: Security for Reliable Swarm Cyber-Physical System
- Courseworks: {Computer Security, Computer Network, Computer Architecture, Operating Systems, Computer Algorithms, Control Engineering, Real-time Embedded Systems}

**Summer Session Visiting Student** — June 2023 - August 2023
UC Berkeley, GPA 4.0 / 4.0 — *Berkeley, California*

- Visited by FGLP (Freshman Global Leadership Program), supported by DGIST

**Completion of Technical Venture Leader Course** — 23 February 2023
DGIST TVA (Technical Venture-leader Academy) — *Daegu, South Korea*

## Publications

**Demo: One Shot All Kill: Building Optimal Attack on Swarm Drones**
*Symposium on Vehicle Security and Privacy (VehicleSec 2024), San Diego, United States*
***Minki Lee***, Gangmin Kim, Jonghyun Kang, Hyunwoo Kim, Jangwon Lee, Hongjun Choi
https://dx.doi.org/10.14722/vehiclesec.2024.25008



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## Awards and Honors

**Special Award:** KIISC President's Award | *2025 ACDC (AI Cyber Defense Contest)* December 2025
- Awarded by the Korea Institute of Information Security & Cryptology (KIISC).
- Selected as a **Finalist** (Team RHCP) after ranking **5th** (student league) in the qualifier.
- *Focus:* Adversarial attacks and defense strategies for AI models (AI for Security).

**Tech Award:** KDIA President's Award | *HackTheDrone 2024* October 2024
- Awarded by the Korea Drone Innovation Association (Team CompSec).
- Ranked **7th** in Finals and **2nd** in the Qualifier.
- *Focus:* Drone protocol vulnerabilities, packet analysis, and RF signal jamming using SDR.

**Grand Prize (DGIST President's Award)** | *DGIST Startup Contest* October 2022
- Awarded for the most innovative business model among pre-startup teams.
- Won **2nd place** overall as a team leader.
- *Focus:* Proposed a Personalized Learning SaaS platform using AI-driven analytics.

**Full National Scholarship** | *DGIST* February 2021 – August 2025
- Awarded full tuition support for continuous academic excellence (8 semesters).

## Experiences

### [Research Experiences]

**Visiting Undergraduate Intern** June 2024 - August 2025
CompSec Lab, POSTECH *Pohang, South Korea*
- Advisor: Seulbae Kim
- Computer Security & CPS Security
- Research in Drone Swarm System Testbed

**Undergraduate Student Researcher** November 2022 – June 2024
Cyber-Physical Security (CPSec) Lab, DGIST *Daegu, South Korea*
- Advisor: Hongjun Choi
- Research about Secure and Robust Operation of Future Mobility (Drone Swarm)
- UGRP (Undergraduate Group Research Project) for Resilient Swarm Drones
- Presented Demo Poster at VehicleSec 2024, San Diego, United States

**Research Internship** December 2023 – January 2024
2024 Winter Internship Program, Department of EECS, DGIST *Daegu, South Korea*
- CSI Lab, Advisor: Kyung-Joon Park
- Research about Secure and Robust Future Mobility System

### [Teaching Experiences]

**Teaching Assistant** February 2026 - June 2026
CSED415: Computer Security *POSTECH CSE*

**Research (Teaching) Assistant** August 2024 - December 2024
CSED199: Freshman Research Participation *POSTECH CSE*

**Undergraduate Teaching Assistant** August 2024 - December 2024
CSE401: Introduction to Database *School of Undergraduate Studies, DGIST*

### [Work Experiences]

**CUOP (Company - University Cooperation Program) Intern** December 2024 – February 2025
Quester *Daegu, South Korea*
- Intern Developer
- Computer Vision AI model (object detection and tracking) for sensor fusion
- Experienced {YOLOv8, BiSeNet, SORT, DeepSORT, SAMURAI, SLAM}
- Product Security Analysis (Reverse Engineering)

**CUOP (Company - University Cooperation Program) Intern** June 2022 – August 2022
BostonAIMS *Seoul, South Korea*
- Intern Developer
- AI model Server Deployment / Development
- Huggingface, Dart / Flutter, Docker along with Firebase Server